

NAMES:
STUDENT NUMBERS:

MODEL CARD

Predicting student's academic success or dropout

Model Details	
1. Person or organization developing model	Polytechnic Institute of Portalegre, Portugal
2. Model date	2021
3. Model version	
4. Model type	Binary classifier through logistic regression
5. Paper or other resource for more information	Link to dataset Link to published research
6. Citation details	
7. License	
8. Contact details for feedback on the model	

Intended Use	
9. Primary intended uses	Early detection of students at risk of dropping out
10. Primary intended users	University staff (student counsellors)
11. Out-of-scope use cases	Keeping students with predicted risk from enrolling (or other resources)

Factors		
Relevant Factors	12. Groups	Students, University staff
	13. Instrumentation & Environment	Student data collected and kept by the university. The data is highly specific to one Portuguese university since the data is only collected there, so the model is not usable in other contexts.
14. Evaluation Factors	- Student grades from the first semester - Student enrolment status (scholarship holder, tuition fees paid, credits taken in the semester) - Social background (e.g.: parents' occupations and qualifications, international student status, displacement, marital status, age at enrolment)	

Metrics	
15. Metrics	Accuracy, Recall for predicted dropout class, Unweighted average of F1 scores for both classes

Evaluation Data	
16. Evaluation Data	Target variable of our dataset

Training Data	
17. Training Data	See link above Our dataset has data about 4424 anonymous students from the years 2018/2019 and 2008/2009, with 35 variables including demographic background and previous education, as well as grades from 2 semesters. The target column is a categorical variable with 3 values: enrolled, graduated or dropped out

Quantitative Analysis	
18. Quantitative Analysis	After pre-processing steps our model reached ~87% accuracy as well as 85% recall for the dropout class

Ethical Considerations	
19. Data/human life/mitigations/risks harms/use cases	This model processes sensitive data containing proxies for wealth and ethnicity e.g., Debtor, Nationality, which poses risks of reinforcing inequality or stigmatizing students. To mitigate this, the system is designed as strictly advisory, relying on a "human in the loop" to validate predictions and prevent automation bias. Risks of discrimination are addressed by using the output solely for offering proactive support, never for punitive administrative decisions. This ensures the model aids student success without compromising fairness or privacy.

Caveats and Recommendations	
20. Evaluation Data	Since of the data is Portuguese, the model may have poor performance or higher error rates when predicting outcomes for international students or ethnic minorities. As counselors intervene and save "at-risk" students, the dropout rate will drop. The historical training data will eventually stop reflecting reality (it will be too pessimistic), requiring the model to be retrained.

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